

Verification Plan Report

Project: AXI-Stream Master VIP - High-Density Enterprise Plan

Version: 1.2

Verification Role: Master

1. UVM Environment Architecture

```
graph TD
    classDef test_layer fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px,color:#7b1fa2;
    classDef test_obj fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px,stroke-dasharray: 5
5,color:#7b1fa2;
    classDef env_layer fill:#e3f2fd,stroke:#1565c0,stroke-width:2px,color:#1565c0;
    classDef env_obj fill:#e3f2fd,stroke:#1565c0,stroke-width:2px,stroke-dasharray: 5
5,color:#1565c0;
    classDef agent_layer fill:#e0f2f1,stroke:#00695c,stroke-width:2px,color:#00695c;
    classDef agent_obj fill:#e0f2f1,stroke:#00695c,stroke-width:2px,stroke-dasharray: 5
5,color:#00695c;
    classDef dut_layer fill:#fff8e1,stroke:#f57f17,stroke-width:2px,color:#f57f17;

    subgraph UVM_Test_Layer [axi_stream_master_test]
        SeqItem[axi_stream_seq_item]:::test_obj
        Seq[axi_stream_master_sequence]:::test_obj
        subgraph Env_Layer [axi_stream_env]
            Scoreboard[axi_stream_scoreboard]:::env_layer
            Env_Cfg[axi_stream_env_config]:::env_obj
            subgraph Agent_Layer [Master Agent (VIP)]
                Sequencer[axi_stream_master_sequencer]:::agent_layer
                Agent_Cfg[axi_stream_master_agent_config]:::agent_obj
                Driver[axi_stream_master_driver]:::agent_layer
                Monitor[axi_stream_master_monitor]:::agent_layer
                Sequencer --> Driver
                Agent_Cfg --> Sequencer
            end
        end
    end

    subgraph Top_Level [Hardware Top]
        VIF[Virtual Interface (axi_stream_if)]:::dut_layer
        DUT[DUT Slave Receiver]:::dut_layer
        VIF <--> DUT
    end

    Driver --- VIF
    Monitor --- VIF
    Monitor --> Scoreboard
```

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2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MST_01	Handshake (Stability)	TVALID Stability until Handshake	Once TVALID is asserted, it must remain HIGH and all signals must be stable until TREADY is asserted. Verification prevents 'glitch' valid or premature data changes during Slave backpressure.
REQ_MST_02	Handshake (Independence)	TVALID Independence from TREADY	Master must not wait for TREADY before asserting TVALID. Verifies that the Master VIP correctly initiates transfers based on internal data availability.
REQ_MST_03	Clock & Reset (Reset Timing)	Deassertion after Reset	TVALID must be LOW during reset and for 1 cycle after reset release. Prevents illegal transactions during hardware initialization.
REQ_MST_04	Packets (Continuous)	Continuous Packet Constraint Enforcement	In Continuous mode, TID/TDEST must not change between TLAST=0 transfers, and all TKEEP bits must be HIGH. Verification ensures the VIP strictly follows high-efficiency streaming rules.
REQ_MST_05	Data Streams (Null Transfers)	TLAST flushing without payload	A transfer with TLAST=1 can have zero data (TKEEP=0). Used for pushing buffered data. Verifies Slave endpoint synchronization.
REQ_MST_06	Byte Qualifiers (Sparse Data)	TKEEP vs TSTRB Interaction	Differentiates between valid information (TSTRB) and bytes that must reach the destination (TKEEP). Verifies correct reconstruction of position bytes at the Receiver.
REQ_MST_07	Interleaving (Multi-Stream)	Per-Transfer TID Switching	Master is permitted to interleave different streams (TIDs) on every cycle. Verifies Interconnect arbitration and Slave reordering/ID handling.
REQ_MST_08	AXI5 Features (Wakeup)	TWAKEUP Assertion Priority	TWAKEUP must be asserted at least 1 cycle before TVALID (recommended). Verifies clock-gating and power-up sequences of the connected IP.

3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MST_001	Directed Handshake: TVALID before TREADY with 1-100 cycle random stall.	REQ_MST_01
TC_MST_002	Directed Handshake: TREADY before TVALID (Receiver Ready mode).	REQ_MST_01
TC_MST_003	Directed Handshake: Zero-wait simultaneous TVALID and TREADY.	REQ_MST_01
TC_MST_004	Data Stability: Random TDATA change attempt during TREADY stall (Expected: No change on bus).	REQ_MST_01
TC_MST_005	Deadlock Check: Hold TREADY LOW for 1000 cycles, verify TVALID is still asserted.	REQ_MST_02

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TC_MST_006	Reset Exit: Verify TVALID is strictly LOW on cycle N+1 after reset deassertion.	REQ_MST_03
TC_MST_007	Reset Mid-Transfer: Toggle ARESETn in the middle of a packet. Verify TVALID reset.	REQ_MST_03
TC_MST_008	Continuous Mode: Send Packet (L=16) with TID change halfway (Negative Test: Should Fail).	REQ_MST_04
TC_MST_009	Continuous Mode: Send Packet with TKEEP=0 halfway (Negative Test: Should Fail).	REQ_MST_04
TC_MST_010	Flush Packet: Handshake with TLAST=1, TVALID=1, TREADY=1, TDATA=X, TKEEP=0.	REQ_MST_05
TC_MST_011	Back-to-Back Flushes: Send three TLAST=1, TKEEP=0 packets consecutively.	REQ_MST_05
TC_MST_012	Sparse Transfer: Random TSTRB/TKEEP masks (e.g., TSTRB=1010, TKEEP=1111).	REQ_MST_06
TC_MST_013	Position Byte: Valid information in byte 0 and 3 only (TSTRB=1001).	REQ_MST_06
TC_MST_014	ID Interleaving: ID0-Trans-ID1-Trans-ID0-Trans-ID2.	REQ_MST_07
TC_MST_015	Dest Interleaving: Switching TDEST on every handshake to verify routing.	REQ_MST_07
TC_MST_016	AXI5 Wakeup: Assert TWAKEUP, wait 1 cycle, then assert TVALID.	REQ_MST_08
TC_MST_017	Wakeup Glitch: Toggle TWAKEUP without TVALID (Recommended: Acceptable but wasteful).	REQ_MST_08
TC_MST_018	Max Throughput: Sustain TVALID/TREADY HIGH for 1000 contiguous cycles.	REQ_MST_01
TC_MST_019	Random Stall Profile: Driver injects 1-5 cycle bubbles between transfers.	REQ_MST_01
TC_MST_020	Full ID Cross: Cycles through all 256 TID values.	REQ_MST_07
TC_MST_021	Packet Boundary Alignment: Test packets with length equal to word size and offset.	REQ_MST_04
TC_MST_022	Negative: Assert TLAST while TVALID is LOW (Expected: No effect).	REQ_MST_01
TC_MST_023	Negative: Change TDATA while TVALID is HIGH and TREADY is LOW.	REQ_MST_01
TC_MST_024	Packet Preservation: Verify TLAST count at Receiver matches Master assertion count.	REQ_MST_04
TC_MST_025	Arbitration Point: Verify Interconnect switches arbitration after TLAST.	REQ_MST_04
TC_MST_026	Large Packet (1KB): Verifies internal buffer handling of the Slave DUT.	REQ_MST_04
TC_MST_027	Small Packet (1 Byte): Corner case for packet logic.	REQ_MST_04
TC_MST_028	Sparse TSTRB with TLAST: Verify packet boundary on non-fulfilling word.	REQ_MST_06
TC_MST_029	User Sideband: Randomize TUSER on every cycle and verify end-to-end receipt.	REQ_MST_02

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TC_MST_030	Clock Frequency Step: Verify handshake integrity when frequency changes (if supported).	REQ_MST_03
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4. Functional Coverage Groups

Group Name	Description
cg_handshake_transitions	Binning of TVALID/TREADY edge combinations.
cg_packet_lengths	Auto-binning of lengths 1 to 1024 bytes.
cg_byte_masks	Cross of TKEEP and TSTRB for data-width 4 bytes.
cg_id_streams	Hit count for every unique TID/TDEST pair.

5. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input	1	Clock.
ARESETn	Input	1	Reset.
TVALID	Output	1	Valid.
TREADY	Input	1	Ready.
TDATA	Output	32	Data.
TKEEP	Output	4	Keep.
TSTRB	Output	4	Strobe.
TLAST	Output	1	Last.
TID	Output	8	ID.
TDEST	Output	8	Dest.
TUSER	Output	4	User.
TWAKEUP	Output	1	Wakeup (AXI5).

6. Verification Environment Structure

```
axi_stream_master_enterprise_tb/  
|-- master_agent/  
|   |-- axi_stream_master_driver.sv  
|   |-- axi_stream_master_monitor.sv  
|   |-- axi_stream_master_sequencer.sv  
|   |-- axi_stream_master_agent_config.sv  
|-- env/  
|   |-- axi_stream_scoreboard.sv  
|   |-- axi_stream_coverage_collector.sv  
|   |-- axi_stream_env_config.sv
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```
|-- sequences/
|   |-- axi_stream_master_base_seq.sv
|   |-- axi_stream_master_random_seq.sv
|   |-- axi_stream_master_reset_seq.sv
|   |-- axi_stream_master_interleave_seq.sv
|-- top/
    |-- axi_stream_tb_top.sv
    |-- axi_stream_test_pkg.sv
```

7. SystemVerilog Implementation

axi_stream_master_sequencer.sv (Multi-Stream Support)

```
class axi_stream_master_sequencer extends uvm_sequencer #(axi_stream_seq_item);
    `uvm_component_utils(axi_stream_master_sequencer)
    // Advanced arbitration logic for TID/TDEST interleaving goes here
endclass
```